02443 Stochastic Simulation

Day 3: Ferry example

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Case study: Ferry operations between Rødby and Puttgarden

This lecture is based on a real consultancy project concerning ferry operations between Rødby and Puttgarden.

Two ferry operators wish to use the same harbour facilities:

- One operator is already established.
- A potential competitor seeks access to the route.

Key challenge

The new operator should not significantly disrupt the operations of the existing operator. However, acceptable levels of disruption are subject to legal and operational constraints.

Why simulation?

Conducting real-world experiments would be impractical and potentially very costly. Instead, we can build a simulation model and investigate different operating scenarios before any changes are implemented.

Available data

- Sailing times
- Loading and unloading times
- Harbour capacity constraints
- Safety and operational regulations

Elements of a discrete event simulation

Before implementing a simulation model, we must identify the key components of the system.

Key components

- Simulation clock
- State variables
- Event list
- Statistical quantities

Simulation clock

Tracks the current simulation time.

State variables

Describe the current state of the system.

- Is the harbour channel occupied?
- Which berths are available?
- Where is each ferry currently located?

Elements of a discrete event simulation

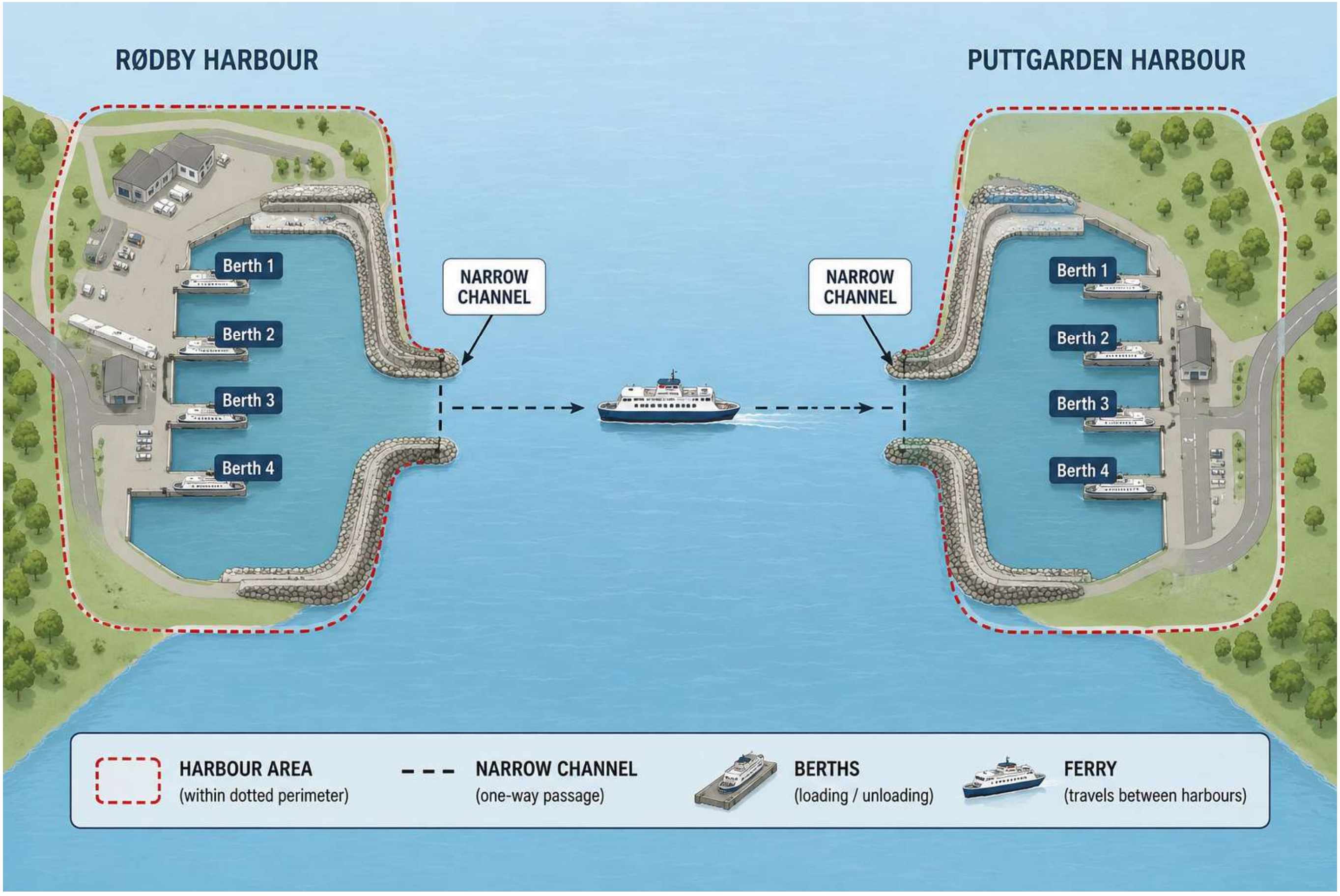
Event list
Contains all scheduled future events.

- Ferry arrives at/departs harbour.
- Ferry arrives at/departs berth.
- Loading completed.
- Ferry ready to sail.
- Ferry arrives at/departs the channel.

Statistical accumulators
Collect performance measures of interest.

- Delays experienced by ferries.
- Utilization of berths and channels.
- Number of delayed departures.

Illustration



Credit: ChatGPT

Exercise 4

Write a discrete event simulation program for a blocking system, i.e. a system with m service units and no waiting room. The offered traffic A is the product of the mean arrival rate and the mean service time.

1. The arrival process is modeled as a Poisson process. Report the fraction of blocked customers, and a confidence interval for this fraction. Choose the service time distribution as exponential.

   Parameters: $m = 10$, mean service time is 8 time units, mean time between customers is 1 time unit (corresponding to an offered traffic of 8 Erlang), 10 x 10.000 customers.

   This system is sufficiently simple such that the analytical solution is known. See the last slide for the solution. Verify your simulation program using this knowledge.

Exercise 4

Write a discrete event simulation program for a blocking system, i.e. a system with m service units and no waiting room. The offered traffic A is the product of the mean arrival rate and the mean service time.

2. The arrival process is modeled as a renewal process using the same parameters as in Part 1 when possible. Report the fraction of blocked customers, and a confidence interval for this fraction for at least the following two cases:

   a) Experiment with Erlang distributed inter arrival times The Erlang distribution should have a mean of 1 time unit.

   b) Hyperexponential inter-arrival times. The parameters for the hyperexponential distribution should be $p_1 = 0.8, \lambda_1 = 0.8333, p_2 = 0.2, \lambda_2 = 5.0$.

Exercise 4

Write a discrete event simulation program for a blocking system, i.e. a system with m service units and no waiting room. The offered traffic A is the product of the mean arrival rate and the mean service time.

3. The arrival process is again a Poisson process like in Part 1. Experiment with different service time distributions with the same mean service time and m as in Part 1 and Part 2.

   a) Constant service time

   b) Pareto distributed service times with at least $k = 1.05$ and $k = 2.05$.

   c) Choose one or two other distributions.

Exercise 4

Write a discrete event simulation program for a blocking system, i.e. a system with m service units and no waiting room. The offered traffic A is the product of the mean arrival rate and the mean service time.

5. Compare confidence intervals for Parts 1, 2, and 3 then interpret and explain differences if any.

Exact solution

Consider a system with a Poisson arrival process with intensity λ and a general service time distribution with mean service time s. Define then the offered traffic $A = \lambda s$.

The Erlang's B-formula applies:

$$B = P(m) = \frac{\frac{A^m}{m!}}{\sum_{i=0}^{m} \frac{A^i}{i!}}.$$